

Robust SDF

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1 Methodology

1.1 Model setup

Following the framework of Hansen and Jagannathan (1991), we consider a Stochastic Discount Factor (SDF) that resides in the linear span of excess returns:

$$M_t = 1 - b'_{t-1}(R_t - \mathbb{E}[R_t]), \quad (1)$$

where the $N \times 1$ vector of SDF loadings, b_{t-1} , is determined by the conditional pricing equation:

$$\mathbb{E}_{t-1}[M_t R_t] = 0. \quad (2)$$

Characteristics-based asset pricing models typically parameterize the SDF loadings as:

$$b_{t-1} = Z_{t-1}b, \quad (3)$$

where Z_{t-1} denotes an $N \times K$ matrix of observable asset characteristics and b is a $K \times 1$ vector of time-invariant coefficients.

Substituting Eq. (3) into Eq. (1) yields an SDF expressed in the linear span of K characteristic-managed portfolios, or factors, defined as $F_t = Z'_{t-1}R_t$. Consequently, the SDF can be rewritten as:

$$M_t = 1 - b'(F_t - \mathbb{E}[F_t]). \quad (4)$$

Following the recent literature on characteristic-based factor models (e.g., Kozak et al. (2018) and Kozak et al. (2020)), we focus our estimation on the unconditional pricing identity:

$$\mathbb{E}[M_t F_t] = 0. \quad (5)$$

In this setup, the factors F_t play a dual role: they serve as the test assets whose return we aim to explain, and simultaneously, they represent the candidate factors that constitute the SDF.

1.2 Estimation

To estimate the SDF coefficient b , a standard approach is to replace population moments with their sample counterparts and solve the sample versions of Eq. (4) and Eq. (5). This yields the estimator:

$$\hat{b} = \Sigma_0^{-1} \mu_0, \quad (6)$$

where $\mu_0 \equiv \frac{1}{T} \sum_{t=1}^T F_t$ and $\Sigma_0 \equiv \frac{1}{T} \sum_{t=1}^T (F_t - \mu_0)(F_t - \mu_0)'$ represent the sample mean and covariance matrix of the factors, respectively.

However, given the high dimensionality of the potential factor space (i.e., large K), sample moments often serve as noisy proxies for their population counterparts, making the estimator in Eq. (6) susceptible to spurious overfitting. To enhance the model's out-of-sample performances, we propose a robust estimation framework that allows for potential deviations between population and sample moments. Specifically, we obtain a robust SDF coefficient, $\hat{b}_{\text{robust}}$, by minimizing the pricing error under the least-favorable distribution within a prescribed ambiguity set.

To motivate our approach, we first note that the standard estimator in Eq. (6) can be derived from the following quadratic loss minimization problem:

$$\min_b \quad \|\mu_0 - \Sigma_0 b\|^2, \quad (7)$$

where $\|\mu_0 - \Sigma_0 b\|$ represents the sample pricing error. In the spirit of Hansen and Jagannathan (1997), this distance measures the discrepancy between the candidate SDF and the space of valid pricing kernels. Following Delage and Ye (2010), we assume that the true moments (μ, Σ) of F_t reside within a neighborhood of the sample moments, defined by the following ambiguity set:

$$\begin{aligned} (\mu - \mu_0)' \Sigma_0^{-1} (\mu - \mu_0) &\leq \gamma_1, \\ (1 - \gamma_2) \Sigma_0 &\preceq \Sigma \preceq (1 + \gamma_2) \Sigma_0, \end{aligned} \quad (8)$$

where $\Sigma = \mathbb{E}[(F_t - \mu_0)(F_t - \mu_0)']$. We then formulate the estimation of b as a minimax problem, minimizing the worst-case pricing error over the ambiguity set:

$$\min_b \max_{\mu, \Sigma} \quad \|\mu - \Sigma b\|^2. \quad (9)$$

Equation (9) characterizes a Distributionally Robust Optimization (DRO) problem. Here, the investor or econometrician is uncertain about the true distribution of factors F_t , knowing only that its first two moments lie within the uncertainty set specified in (8). Consequently, $\hat{b}_{\text{robust}}$ identifies the SDF coefficients that remain performant even under the most adverse distributional realization of the factor returns.

To provide further economic intuition, we demonstrate that a specific case of this DRO framework is equivalent to a regularized regression. Consider the following reformulated problem:

$$\min_b \max_{\mu, \Sigma} \quad (\mu - \Sigma b)' \Sigma_0^{-1} (\mu - \Sigma b), \quad (10)$$

which employs the Mahalanobis distance to characterize the pricing error. The following theorem establishes the connection between our robust estimator and the square-root Lasso:

Theorem 1. *Under the moment constraints in (8), the solution to the robust optimization problem in (10) is equivalent to the solution of the following Square-root Lasso problem:*

$$\min_{\tilde{b}} \quad \|\mu_0 - \Sigma_0 \tilde{b}\|_{\Sigma_0^{-1}} + \gamma_2 \|\tilde{b}\|_{\Sigma_0} \quad (11)$$

Proof of Theorem 1. By invoking the dual representation of the squared Mahalanobis distance, the minimax optimization problem in (10) can be reformulated as:

$$\min_b \max_{\mu, \Sigma} \max_y \{2y'(\mu - \Sigma b) - \|y\|_{\Sigma_0}^2\}. \quad (12)$$

Since the constraints for μ and Σ in (8) are independent, we can rearrange the nested maximization. Specifically, the problem can be decomposed as follows:

$$\min_b \left\{ \max_y \left[\max_{\mu} (2y'\mu) + \max_{\Sigma} (-2y'\Sigma b) - \|y\|_{\Sigma_0}^2 \right] \right\}. \quad (13)$$

Under the ambiguity set defined in (8), the inner optimizations over μ and Σ yield the following closed-form solutions:

$$\max_{\mu} (2y'\mu) = 2y'\mu_0 + 2\sqrt{\gamma_1}\|y\|_{\Sigma_0}, \quad \text{and} \quad \max_{\Sigma} (-2y'\Sigma b) = \gamma_2\|b\|_{\Sigma_0}\|y\|_{\Sigma_0} - y'\Sigma_0 b. \quad (14)$$

Substituting these results back into the objective function, we obtain:

$$\begin{aligned} & \min_b \left\{ \max_y \left[(2y'\mu_0 + 2\sqrt{\gamma_1}\|y\|_{\Sigma_0}) + 2\gamma_2\|b\|_{\Sigma_0}\|y\|_{\Sigma_0} - 2y'\Sigma_0 b - \|y\|_{\Sigma_0}^2 \right] \right\} \\ &= \min_b \left\{ \max_y \left[2y'(\mu_0 - \Sigma_0 b) + 2(\sqrt{\gamma_1} + \gamma_2\|b\|_{\Sigma_0})\|y\|_{\Sigma_0} - \|y\|_{\Sigma_0}^2 \right] \right\}. \end{aligned} \quad (15)$$

To simplify the expression, we perform a change of variables. Let $u = \Sigma_0^{1/2}y$ and define the auxiliary variables $\tilde{v} = \Sigma_0^{-1/2}(\mu_0 - \Sigma_0 b)$ and $C = \sqrt{\gamma_1} + \gamma_2\|b\|_{\Sigma_0}$. The inner problem then reduces to:

$$\begin{aligned} & \max_u \quad 2u'\tilde{v} + 2C\|u\| - \|u\|^2 \\ &= (\|\tilde{v}\| + C)^2. \end{aligned} \quad (16)$$

Taking the square root and substituting the definitions of $\tilde{v}$ and C , the minimization problem is equivalent to:

$$\begin{aligned} & \min_b \quad \|\tilde{v}\| + C \\ &= \min_b \quad \|\mu_0 - \Sigma_0 b\|_{\Sigma_0^{-1}} + \gamma_2\|b\|_{\Sigma_0} + \sqrt{\gamma_1}. \end{aligned} \quad (17)$$

In this problem, γ_1 has no effect on the solution. Then the original optimization problem is equivalent to :

$$\min_{\tilde{b}} \quad \|\mu_0 - \Sigma_0 \tilde{b}\|_{\Sigma_0^{-1}} + \gamma_2\|\tilde{b}\|_{\Sigma_0} \quad (18)$$

This completes the proof. $\square$

While the objective function in (10) utilizes second moments centered at the sample mean μ_0 , a theoretically consistent pricing error should be constructed using the second moment centered at the perturbed mean μ . To maintain such internal consistency within the ambiguity set, we propose a more rigorous formulation of the robust optimization problem:

$$\min_b \max_{\mu, \Sigma} \quad (\mu - \mathbb{E}[(F_t - \mu)(F_t - \mu')]'b)' \Sigma_0^{-1} (\mu - \mathbb{E}[(F_t - \mu)(F_t - \mu')]'b), \quad (19)$$

where the expectation in the second moment is evaluated under the endogenous mean μ within the ambiguity set. This increased rigor, however, comes at the cost of analytical tractability. Unlike the simplified case in Theorem 1, the optimization in (19) does not admit a direct closed-form equivalent and therefore necessitates the use of numerical solution methods.

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